Table 11: **Results for code-augmentation for 2.8B parameter models.** Models trained on a mix of natural language (C4) and Python (The Stack). Scores are normalized averages of 0-5 few-shots and reported as percentages. We report mean/std. err. across five different models, each trained with a different random seed.

	% of Python pre-training data (rest is C4)						% of Python pre-training data (rest is OSCAR)					
Dataset (↓)	0	10	20	30	40	50	0	10	20	30	40	50
ANLI R1	0.4 ± 1.6	-1.5	-0.9	-1.0	-0.7	-2.4	-0.3 ± 0.5	0.0	-0.6	-1.6	-2.4	-1.7
ANLI R2	0.9 ± 0.4	0.7	0.0	0.1	-0.1	0.1	1.0 ± 1.0	1.2	-0.1	-0.0	0.0	0.8
ANLI R3	1.7 ± 0.5	0.6	-0.7	-0.2	0.4	0.0	0.4 ± 0.8	-0.4	-0.2	-1.7	-0.8	-0.5
ARC-Challenge	1.6 ± 1.0	4.2	1.7	1.5	0.2	-0.2	-1.4 ± 0.8	-0.7	-1.4	-3.4	-2.3	-3.1
ARC-Easy	44.5 ± 0.5	46.4	46.5	45.4	43.6	42.7	39.7 ± 0.3	39.8	38.7	39.1	37.3	37.6
BoolQ	18.8 ± 3.4	15.7	19.0	13.4	16.0	4.4	12.8 ± 4.4	3.3	12.5	10.6	5.8	8.5
CB	20.0 ± 4.7	22.8	10.7	20.5	17.4	15.2	19.7 ± 5.1	14.7	15.6	19.6	22.8	17.0
COPA	49.7 ± 3.5	46.3	49.3	46.3	42.7	40.0	42.7 ± 2.2	42.7	41.0	42.7	35.7	38.0
HellaSwag	24.7 ± 0.3	24.1	23.3	22.3	21.9	20.9	16.3 ± 0.1	15.7	15.9	15.5	15.1	13.7
PiQA	47.9 ± 0.6	46.9	47.7	45.1	46.2	45.5	41.2 ± 0.7	41.6	39.9	40.5	38.8	38.6
RTE	5.1 ± 4.0	8.8	7.7	5.1	7.8	10.8	3.9 ± 1.1	2.2	4.3	1.1	3.7	-1.7
SciQ	83.2 ± 0.6	83.3	85.3	84.8	83.2	83.7	83.2 ± 0.6	82.4	83.5	82.8	83.3	83.2
StoryCloze 2016	58.7 ± 0.2	59.3	57.9	56.9	56.5	56.0	52.8 ± 0.3	52.0	52.2	52.0	51.8	50.9
WinoGrande XL	11.6 ± 0.8	13.0	10.7	9.3	8.2	9.6	5.8 ± 0.9	3.2	5.6	5.8	4.6	3.9
E2E NLG	17.0 ± 1.4	19.8	21.1	20.2	22.1	21.0	20.3 ± 0.3	21.9	20.7	20.5	20.7	21.1
XSUM	2.4 ± 0.1	2.7	2.0	2.2	2.0	2.3	3.0 ± 0.1	2.8	3.1	3.4	3.1	2.9
WebNLG EN	5.3 ± 0.1	9.1	8.0	8.5	8.5	9.1	8.8 ± 0.4	8.7	9.6	9.1	8.7	9.4
WikiLingua EN	3.0 ± 0.1	3.2	3.2	3.6	3.3	3.7	2.9 ± 0.1	3.3	3.6	3.5	3.4	3.5
bAbI	0.0 ± 0.0	4.6	14.2	14.2	14.8	15.1	15.5 ± 1.0	16.6	17.2	17.2	17.7	15.9
Average	20.9 ± 0.4	21.6	21.4	21.0	20.7	19.9	19.4 ± 0.5	18.5	19.0	18.8	18.3	17.8
Average (without bAbI)	22.0 ± 0.5	22.5	21.8	21.3	21.1	20.1	19.6 ± 0.5	18.6	19.1	18.9	18.3	17.9

N Filtering Procedure

Perplexity filtering We follow the approach of [54] to perform perplexity filtering and reuse their artifacts - a SentencePiece tokenizer [52] and a KenLM 5-gram language model [37] trained on Wikipedia introductions and available to download from their repository.[4] We compute the model's perplexity on all OSCAR and C4 samples and only select samples that fall within a certain percentile threshold. For example, to select the top 25%, we only select samples with perplexity lower than the 25th percentile. Figure 18 provides a visual representation of perplexity distribution for respective datasets, highlighting the relevant percentile thresholds.

Deduplication We perform deduplication leveraging the suffix array-based approach proposed by Lee et al. [55]. We remove any document with at least a 100-character span overlapping with any other document in the corpus. We deduplicate the full C4 dataset. In the case of OSCAR, the memory requirements of the deduplication procedure make performing the full dataset deduplication infeasible. Instead, we select a 25% subset of the full OSCAR and build a suffix array for this subset. We experiment with leveraging the 25% OSCAR suffix array in two ways. First, we deduplicate the selected subset. This is very strict and preserves less than 5% of the full OSCAR. Subsequently, we use the 25% suffix array to deduplicate the full OSCAR, i.e. we remove any document which has at least a 100-character span overlapping with the 25% subset we selected. This is more permissive and allows us to preserve 31% of the original dataset. We refer to the latter as *expanded* in Table 12 and it is used for the training of the 4.2 billion parameter model in Table 14, while the smaller deduplicated version of OSCAR is used for the 2.8 billion parameter model.

ROOTS filter In addition, we benchmark with the filtering procedure from the ROOTS corpus [54]. It applies the following set of filters:

- Discarding documents with too few words
- Discarding documents with overly repeated character- and word-n-grams
- Discarding documents with too many special characters

[4] https://github.com/bigscience-workshop/data-preparation/tree/main/preprocessing/training/01b_oscar_cleaning_and_filtering

- Discarding documents with too few grammatical function words (e.g. “of”, “and”)
- Discarding documents with too many flagged words
- Discarding documents with a low `fasttext` language identification score
- Perplexity filtering

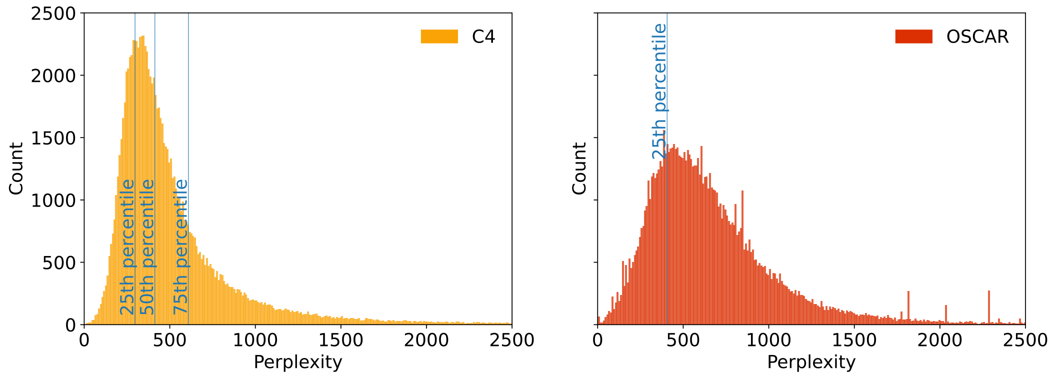

Figure 18: Perplexity histograms for respective datasets. For demonstration purposes, we use 100,000 random samples of each dataset.

Table 12: **Sizes of filtered datasets.**

Base Dataset	Filter	Tokens after filtering
C4	Deduplication	21 billion
C4	Perplexity Top 25%	44 billion
C4	Perplexity Top 50%	89 billion
C4	Perplexity 25-75%	89 billion
OSCAR	Deduplication	9 billion
OSCAR	Deduplication-expanded	94 billion
OSCAR	Perplexity Top 25%	80 billion
OSCAR	ROOTS	99 billion

O Detailed Filtering Results

In Table 13, we report detailed perplexity filtering results on C4 and OSCAR. For C4, perplexity filtering is only effective at 4.2B parameters. Meanwhile, for OSCAR, which is noisier than C4, perplexity filtering seems effective both for 2.8B and 4.2B parameters. Table 14 contains deduplication results and results for the ROOTS filter. Deduplication does not improve downstream performance for C4 while being effective for OSCAR which has significantly more noise. Applying the ROOTS filter on OSCAR is not better than the unfiltered OSCAR on our benchmark, but might have other beneficial effects, such as reducing obscenity, templated messages, or repetition, depending on the final use case.

Table 13: **Results for perplexity-filtering.** The training data is perplexity filtered according to the given percentile, e.g. 25% corresponds to training on the top 25% percent of examples with the lowest perplexity. The resulting dataset sizes are in Table 12. The data is repeated until it matches 55B tokens for 2.8B parameter and 84B tokens for 4.2B parameter models. Scores are normalized averages of 0-5 few-shots and reported as percentages. For unfiltered models, we report mean/std. err. across five different models, each trained with a different random seed.

Training Data	C4							OSCAR			
Parameters	2.8B			4.2B				2.8B		4.2B	
Percentile	All	25%	50%	All	25%	50%	25-75%	All	25%	All	25%
ANLI R1	0.4 ± 1.6	-0.1	0.9	-0.5 ± 1.4	-0.0	-0.7	-0.8	-0.3 ± 0.5	-0.4	-0.4 ± 1.2	-2.2
ANLI R2	0.9 ± 0.4	-0.2	-0.7	0.0 ± 1.3	-0.4	-0.0	1.1	1.0 ± 1.0	1.7	1.0 ± 0.9	0.7
ANLI R3	1.7 ± 0.5	0.5	1.4	0.7 ± 0.5	0.7	2.9	0.4	0.4 ± 0.8	1.7	1.2 ± 0.5	2.1
ARC-Challenge	1.6 ± 1.0	3.3	2.9	4.2 ± 1.6	10.2	9.3	7.9	-1.4 ± 0.8	3.3	1.8 ± 0.8	6.3
ARC-Easy	44.5 ± 0.5	47.3	47.7	48.1 ± 4.8	55.8	53.7	51.0	39.7 ± 0.3	46.8	45.7 ± 0.6	51.8
BoolQ	18.8 ± 3.4	17.1	17.7	22.4 ± 3.3	27.7	23.5	24.5	12.8 ± 4.4	11.8	12.4 ± 5.9	22.2
CB	20.0 ± 4.7	16.1	13.8	9.3 ± 16.6	24.6	22.3	12.5	19.7 ± 5.1	17.0	23.9 ± 3.8	20.1
COPA	49.7 ± 3.5	55.7	56.0	55.3 ± 3.8	60.7	66.0	61.0	42.7 ± 2.2	44.0	41.1 ± 3.0	49.3
HellaSwag	24.7 ± 0.3	24.7	26.0	29.4 ± 1.3	30.7	32.7	33.1	16.3 ± 0.1	19.0	21.0 ± 0.2	23.3
PiQA	47.9 ± 0.6	43.4	45.8	48.8 ± 3.8	47.9	52.2	52.1	41.2 ± 0.7	38.3	45.0 ± 0.6	44.4
RTE	5.1 ± 4.0	5.7	7.3	6.9 ± 3.1	11.9	2.2	10.3	3.9 ± 1.1	-1.2	2.2 ± 4.3	7.0
SciQ	83.2 ± 0.6	82.4	82.8	86.3 ± 1.1	88.6	87.4	88.4	83.2 ± 0.6	84.0	86.3 ± 0.6	86.5
StoryCloze 2016	58.7 ± 0.2	61.1	61.2	62.8 ± 0.5	65.5	65.6	65.1	52.8 ± 0.3	57.9	57.2 ± 0.6	60.2
WinoGrande XL	11.6 ± 0.8	15.3	14.3	18.7 ± 1.0	24.9	22.3	18.7	5.8 ± 0.9	9.7	10.1 ± 1.0	14.8
E2E NLG	17.0 ± 1.4	16.1	16.8	17.9 ± 0.7	18.8	17.8	19.2	20.3 ± 0.3	19.5	21.6 ± 0.7	22.6
XSUM	2.4 ± 0.1	2.6	3.0	3.0 ± 0.3	3.9	3.2	3.0	3.0 ± 0.1	3.2	3.7 ± 0.2	2.7
WebNLG EN	5.3 ± 0.1	4.8	5.1	5.6 ± 0.3	5.4	5.7	5.2	8.8 ± 0.4	6.9	9.3 ± 0.5	10.6
WikiLingua EN	3.0 ± 0.1	3.2	3.3	3.6 ± 0.2	3.4	3.5	3.4	2.9 ± 0.1	3.4	4.0 ± 0.1	3.8
bAbI	0.0 ± 0.0	0.0	0.0	0.0 ± 0.0	0.0	0.0	0.0	15.5 ± 1.0	14.5	19.3 ± 1.0	17.2
Average	20.9 ± 0.4	21.0	21.3	22.2 ± 1.4	25.3	24.7	24.0	19.4 ± 0.5	20.1	21.4 ± 0.5	23.3

Table 14: **Results for filtering with deduplication and the ROOTS filters.** The resulting dataset sizes are in Table 12. The data is repeated until it matches 55B tokens for 2.8B parameter and 84B tokens for 4.2B parameter models. Scores are normalized averages of 0-5 few-shots and reported as percentages. For unfiltered models we report mean/std. err. across five different models, each trained with a different random seed.

Training Data	C4				OSCAR					
Parameters	2.8B parameters		4.2B parameters		2.8B parameters			4.2B parameters		
Method	All	Dedup.	All	Dedup.	All	Dedup.	ROOTS	All	Dedup.-exp.	ROOTS
ANLI R1	0.4 ± 1.6	-0.2	-0.5 ± 1.4	-0.8	-0.3 ± 0.5	-2.1	-1.7	-0.4 ± 1.2	-1.8	1.2
ANLI R2	0.9 ± 0.4	1.1	0.0 ± 1.3	-0.1	1.0 ± 1.0	2.0	0.7	1.0 ± 0.9	-0.5	-0.3
ANLI R3	1.7 ± 0.5	1.8	0.7 ± 0.5	0.4	0.4 ± 0.8	0.4	0.2	1.2 ± 0.5	0.8	-0.3
ARC-Challenge	1.6 ± 1.0	0.6	4.2 ± 1.6	3.9	-1.4 ± 0.8	2.6	-0.9	1.8 ± 0.8	6.8	0.6
ARC-Easy	44.5 ± 0.5	43.0	48.1 ± 4.8	46.8	39.7 ± 0.3	44.6	42.3	45.7 ± 0.6	51.0	47.1
BoolQ	18.8 ± 3.4	1.5	22.4 ± 3.3	2.2	12.8 ± 4.4	3.4	13.4	12.4 ± 5.9	13.0	7.0
CB	20.0 ± 4.7	0.4	9.3 ± 16.6	0.9	19.7 ± 5.1	25.4	14.3	23.9 ± 3.8	25.0	28.1
COPA	49.7 ± 3.5	57.0	55.3 ± 3.8	60.0	42.7 ± 2.2	47.3	37.7	41.1 ± 3.0	55.3	43.0
HellaSwag	24.7 ± 0.3	25.1	29.4 ± 1.3	30.7	16.3 ± 0.1	22.8	17.6	21.0 ± 0.2	26.3	22.4
PiQA	47.9 ± 0.6	49.1	48.8 ± 3.8	53.4	41.2 ± 0.7	45.1	41.9	45.0 ± 0.6	48.5	46.3
RTE	5.1 ± 4.0	3.2	6.9 ± 3.1	0.1	3.9 ± 1.1	6.1	5.8	2.2 ± 4.3	1.1	8.9
SciQ	83.2 ± 0.6	80.4	86.3 ± 1.1	82.2	83.2 ± 0.6	82.6	83.1	86.3 ± 0.6	88.5	86.4
StoryCloze 2016	58.7 ± 0.2	61.8	62.8 ± 0.5	65.2	52.8 ± 0.3	58.1	54.3	57.2 ± 0.6	61.6	58.6
WinoGrande XL	11.6 ± 0.8	13.3	18.7 ± 1.0	19.7	5.8 ± 0.9	12.7	5.6	10.1 ± 1.0	16.2	11.0
E2E NLG	17.0 ± 1.4	15.6	17.9 ± 0.7	14.2	20.3 ± 0.3	20.5	20.5	21.6 ± 0.7	2.4	22.6
XSUM	2.4 ± 0.1	2.1	3.0 ± 0.3	2.5	3.0 ± 0.1	3.2	3.1	3.7 ± 0.2	4.6	3.8
WebNLG EN	5.3 ± 0.1	4.3	5.6 ± 0.3	4.4	8.8 ± 0.4	7.4	7.4	9.3 ± 0.5	9.7	9.4
WikiLingua EN	3.0 ± 0.1	3.2	3.6 ± 0.2	3.2	2.9 ± 0.1	3.0	3.1	4.0 ± 0.1	4.3	4.0
bAbI	0.0 ± 0.0	0.0	0.0 ± 0.0	0.0	15.5 ± 1.0	17.2	14.3	19.3 ± 1.0	21.1	18.0
Average	20.9 ± 0.4	19.1	22.2 ± 1.4	20.5	19.4 ± 0.5	21.2	19.1	21.4 ± 0.5	22.8	22.0